

DESIGN PACK | NO FIELDWORK

Primary research design pack: Medicaid coverage criteria for obesity drugs

Module F of the incretin access-and-value project: screeners, discussion guide, stimulus boards and a discrete choice experiment design. Nothing here has been fielded.

Client	A fictional market-access team at an unnamed manufacturer (illustration only)
Prepared by	Erick Kiprotich Yegon, epidemiologist and data scientist
Date	October 2026
Contact	linkedin.com/in/erickyegon keyegon@gmail.com
Status	Design only: no interviews, screeners or surveys have been fielded; no respondent data exist.

Public aggregate data; no company affiliation or endorsement. I used AI tools to help write code and documentation; the study design, methods and conclusions are my own, and I verified all results.

Contents

1. Brief (fictional access team)	3
2. Screeners	3
2.1 Prescriber screener (physician, nurse practitioner, physician assistant)	3
2.2 Payer screener (Medicaid pharmacy director, managed-care organization medical director)	4
3. Discussion guide (60 minutes)	4
4. Stimulus (fictional)	5
4.1 Coverage-policy board (one page)	6
4.2 Concept board: a once-daily oral option (fictional)	6
5. Discrete choice experiment design	6
5.1 Efficient design	7
5.2 Sample size	8
6. Analysis framework and reporting outline	9
7. Illustrative recommendation slides	10
Illustrative slide 1. Coverage works through managed care, so test the criteria that managed-care organizations apply	10
Illustrative slide 2. Coverage can end quickly; ask what would sustain it	10
Illustrative slide 3. The budget is the lever payers describe; test which assumptions they accept	10
8. Ethics, data and limits	10

Warning

Everything here is a design. No interviews, screeners or surveys have been fielded and no respondent data exist. The “access team” below is **fictional** and the stimulus boards are **fictional illustrations**. Numbers marked *secondary data* come from the public-data analyses (Modules A to E) and are linked to [analysis/outputs/key_numbers.csv](#). Public aggregate data, no company affiliation or endorsement.

1. Brief (fictional access team)

Commissioned by: a fictional market-access team at an unnamed manufacturer (illustration only). **Question:** how do Medicaid prescribers and payers weigh the criteria of an obesity-drug coverage policy, and which criteria would they trade off?

Business question	Why the secondary data raise it
Where does coverage actually operate: fee-for-service or managed care?	Coverage is delivered through managed-care programs in South Carolina and Rhode Island, so a fee-for-service-only estimate is not reliable (Module C, sensitivity 12, deviations.md item 6).
How much do prior authorization (PA) criteria matter?	Prior authorization is documented in 9 of 10 of the covering states and BMI criteria differ (BMI 30 or more; BMI 27 to 29 only with a weight-related condition), but Module C cannot separate PA strictness from coverage itself.
What happens when coverage ends?	After coverage ended in California and Pennsylvania, prescriptions per enrollee fell by –67.2% and –90.8% in one quarter (descriptive, preliminary data, Module C).
Is the effect fast or slow?	The effect rose from 1.3 prescriptions per 1,000 enrollees per quarter in the first quarter to 17.2 after eight (Module C, secondary data), partly reflecting market growth.
What would a payer need to see to keep coverage?	Five-year net cost of 161.3 million dollars for 1 million enrollees in the central case, with a range of 55.6 to 369.4 million across scenarios, mostly because rebates are unknown (Module E, secondary data).

Deliverable: a ranked view of the coverage criteria that prescribers and payers will accept, the trade-offs between them, and the messages a field team could use. **Timing and budget:** not set (design only).

2. Screeners

2.1 Prescriber screener (physician, nurse practitioner, physician assistant)

#	Question	Qualify if	Terminate if
S1	Which best describes your role?	Physician (primary care, endocrinology, cardiology), nurse practitioner, physician assistant	Any other role
S2	Do you currently treat patients enrolled in a state Medicaid program?	Yes, 10% or more of patients	No, or under 10%
S3	In the past 3 months, how many adult patients with obesity or overweight did you see for weight management?	20 or more	Fewer than 20

#	Question	Qualify if	Terminate if
S4	Have you prescribed an injectable GLP-1 or GIP/GLP-1 drug for weight management in the past 12 months?	Yes	No
S5	Do you or an immediate family member work for a pharmaceutical company, market-research firm or advertising agency?	No	Yes
S6	Have you taken part in paid market research on obesity or diabetes drugs in the past 3 months?	No	Yes
S7	Which state do you practise in?	Mix of states that covered obesity drugs and states that did not (Module C list)	Quota full

Quotas (n = 30 interviews): primary care physicians 12, nurse practitioners and physician assistants 8, endocrinologists 6, cardiologists 4; at least 10 practising in states that covered obesity drugs under Medicaid and at least 10 in states that did not; at least 8 in a managed-care-dominant program.

2.2 Payer screener (Medicaid pharmacy director, managed-care organization medical director)

#	Question	Qualify if	Terminate if
P1	Your role?	State Medicaid pharmacy director or deputy, MCO medical director, MCO pharmacy director	Any other
P2	Do you influence or decide drug coverage criteria or prior authorization policy?	Yes, decision or formal input	No
P3	Program type?	State fee-for-service program, MCO serving Medicaid, or both	Commercial-only or Medicare-only role
P4	Has your program covered, considered or declined coverage of obesity drugs since 2021?	Covered, considering or declined (any)	Never discussed
P5	Conflicts: employed by or consulting for a manufacturer, pharmacy benefit manager or research firm in the past 12 months?	No	Yes
P6	Willing to discuss criteria and process in general terms, not confidential contract terms?	Yes	No

Quotas (n = 12): state Medicaid pharmacy directors or deputies 6 (3 from states that covered, 3 from states that did not or ended coverage), MCO medical or pharmacy directors 6 (at least 2 in South Carolina or Rhode Island-type managed-care coverage). **Not asked:** confidential rebate or net-price terms.

3. Discussion guide (60 minutes)

Probes are tied to secondary findings; the line shown is the evidence the moderator holds, not a statement to read out.

	Time	Prescriber version	Payer version	Secondary evidence behind the probe
	0 to 5	Consent, role, practice mix	Consent, role, program structure	none
	5 to 15	Current approach to weight management for Medicaid patients; what limits prescribing	How obesity drugs are handled in your program today; who decides	Part D prescribing is mostly primary care (55.8% of 2024 claims; 27.7% nurse practitioners and physician assistants), diabetes and other covered uses, not obesity use (Module D)
	15 to 25	How coverage criteria change your decision: BMI, comorbidity, renewal, step therapy	Which criteria you use and why; what you changed over time	Covering states record BMI 30 or more, with 27 to 29 allowed if a condition is present (coverage.csv); 9 of 10 covering states recorded PA
	25 to 35	How fast patients start and persist after coverage begins	What uptake you saw or expected; how you monitored it	Effect 1.3 in the first quarter, 11.7 at four and 17.2 at eight quarters per 1,000 enrollees (Module C)
	35 to 45	What happens to your patients if coverage stops	Why coverage was ended or paused; what would bring it back	Prescriptions fell -67.2% (California) and -90.8% (Pennsylvania) in one quarter after coverage ended (preliminary, descriptive)
	45 to 55	Stimulus: policy board and oral-option concept board (reactions, not forced choice)	Stimulus: policy board; which criteria you would tighten or loosen	Budget impact: 161.3 million dollars over five years per 1 million enrollees in the central case; range 55.6 to 369.4 million (Module E, assumptions on rebates)
	55 to 60	Wrap-up: one change to make access work	Wrap-up: one condition for sustained coverage	none

Moderator rules: no brand claims beyond label facts saved in [docs/sources/](#); no leading on price; confirm that rebate or net-price terms are out of scope; record verbatims only, no individual identifiers.

4. Stimulus (fictional)

i Note

Fictional stimulus. Wording is illustrative; no real program's policy and no manufacturer claim is shown.

4.1 Coverage-policy board (one page)

FICTIONAL STIMULUS BOARD ILLUSTRATION ONLY		
	Policy A (fictional)	Policy B (fictional)
Who qualifies	BMI 30 or more	BMI 27 or more with at least 1 weight-related condition
Comorbidity	None required	At least 1 weight-related condition
Prior authorization renewal	Every 6 months	Every 12 months
Step therapy	3-month supervised lifestyle program first	None
Patient out-of-pocket cost per month	\$25	\$0
Product class	Injectable GLP-1 or GIP/GLP-1	Injectable GLP-1 or GIP/GLP-1

Questions for the moderator: Which policy would you prefer for your patients (prescriber) or your program (payer)? What would you change first? Which single criterion would you not accept?

4.2 Concept board: a once-daily oral option (fictional)

FICTIONAL STIMULUS BOARD ILLUSTRATION ONLY	
Concept (fictional): “a once-daily oral tablet for chronic weight management.” Taken by mouth, without injections. A prescriber and patient would decide together, as with any chronic weight-management drug. Coverage criteria would follow the program’s existing policy.	

No efficacy, safety or price claims are shown. The moderator may note only label facts already saved in [docs/sources/](#) (the CMS Medicare GLP-1 Bridge pages list oral tablets among covered products: *CMS Medicare GLP-1 Bridge prescriber page, June 2026*). Questions: What would an oral option change in who you would treat? What would it change in coverage criteria or cost to the program?

5. Discrete choice experiment design

Question: how do prescribers (and, with a changed wording, payers) trade off the criteria of a coverage policy for an obesity drug? Respondents see 12 tasks; each task shows two unlabelled policies (A and B) described by five attributes with three levels each, and choose the one they prefer. There is no opt-out in this design; an opt-out version would add one constant and one task type.

Attribute	Level 1	Level 2	Level 3
BMI threshold	27 or more (with a comorbidity)	30 or more	35 or more
Comorbidity requirement	None	At least 1 weight-related condition	At least 2 weight-related conditions
Prior authorization renewal	Every 6 months	Every 12 months	No renewal after initial approval

Attribute	Level 1	Level 2	Level 3
Step therapy	None	3-month supervised lifestyle program first	Trial of a lower-cost agent first
Patient out-of-pocket cost per month	\$0	\$25	\$75

5.1 Efficient design

The design was generated in R with `idefix::Modfed` (modified Fedorov algorithm) from the full factorial of 243 profiles, effects coding, a **zero prior** (no prior information on preferences is assumed; none was invented), 24 tasks split into 2 blocks of 12 (the split with the lowest worst-block error among 600 random splits), seed in the table below. The code is `research_pack/dce_design.R`; the full design is `research_pack/design/dce_design.csv`.

Item	Value
attributes	5
levels per attribute	3,3,3,3,3
alternatives per task	2
tasks in the design	24
blocks	2
tasks per respondent	12
parameters (effects coding)	10
D0-error of the full design (zero prior)	0.0968
D0-error, block 1	0.2297
D0-error, block 2	0.2279
dominated tasks in the final design	0
Johnson-Orme minimum respondents	63
random seed	20267004
package versions	idefix 1.1.0; survival 3.8.6

D-error is the standard efficiency measure (lower is better); here it is the D0-error with a zero prior, and block-level errors are higher because each block has half the tasks. No task in the final design has one policy at least as good as the other on every attribute (dominated tasks: 0). Level balance across the design:

Attribute	Level	Appearances	Share of the attribute's rows (%)
BMI threshold	27 or more (with a comorbidity)	17	35.4
BMI threshold	30 or more	16	33.3
BMI threshold	35 or more	15	31.2
Comorbidity requirement	None	15	31.2
Comorbidity requirement	At least 1 weight-related condition	17	35.4
Comorbidity requirement	At least 2 weight-related conditions	16	33.3
Prior authorization renewal	Every 6 months	16	33.3

Attribute	Level	Appearances	Share of the attribute's rows (%)
Prior authorization renewal	Every 12 months	17	35.4
Prior authorization renewal	No renewal after initial approval	15	31.2
Step therapy	None	17	35.4
Step therapy	3-month supervised lifestyle program first	16	33.3
Step therapy	Trial of a lower-cost agent first	15	31.2
Patient out-of-pocket cost per month	\$0	16	33.3
Patient out-of-pocket cost per month	\$25	16	33.3
Patient out-of-pocket cost per month	\$75	16	33.3

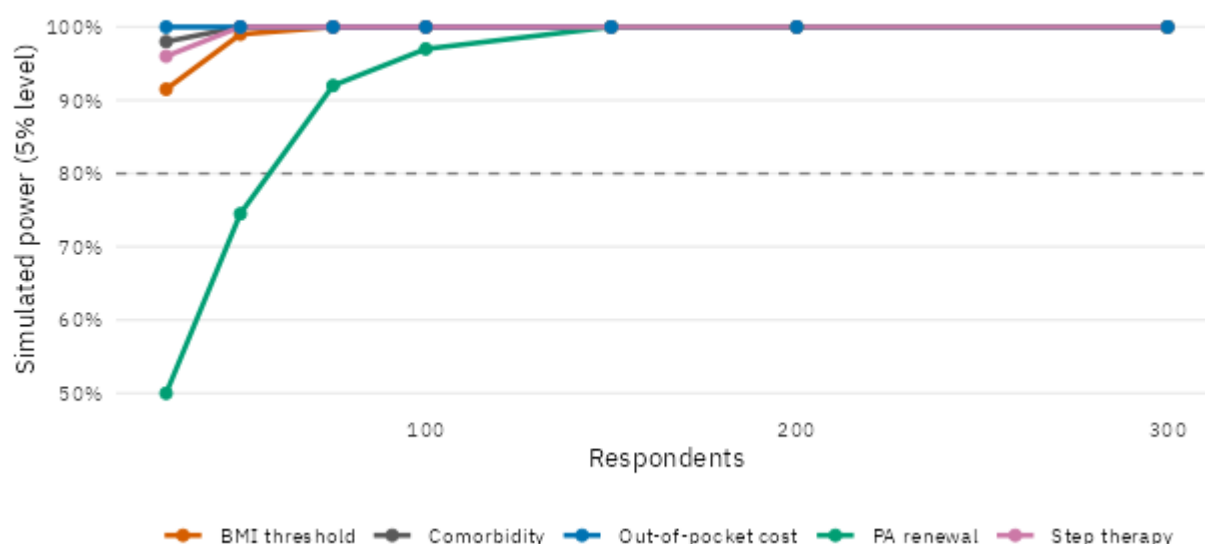
5.2 Sample size

Method 1, Johnson-Orme rule of thumb (as described in [de Bekker-Grob et al. 2015](#) and in [Sawtooth Software's rules of thumb](#)): $N > 500 \cdot c / (t \cdot a)$, where c is the largest number of levels of any attribute (3), t the number of tasks per respondent (12) and a the number of alternatives per task (2): $N > 500 \times 3 / (12 \times 2) = \mathbf{63}$ **respondents** for main effects. Sawtooth's first rule of thumb adds 300 respondents per study and 200 per subgroup to be compared.

Method 2, simulation. For each sample size, 200 simulated studies were run: respondents are randomly assigned to a block, choose between the two policies according to a multinomial logit model with **illustrative true preference weights** (an assumption: adjacent-level utility steps of 0.30 for BMI threshold, 0.40 for comorbidity, 0.20 for renewal interval, 0.35 for step therapy and 0.50 for cost, in logit units, chosen only to make the smallest effect small enough to be demanding), and the model is estimated by conditional logit. Power is the share of simulated studies in which a joint test of each attribute is significant at 5%. These are not estimated from data and would be replaced by pilot estimates.

N	BMI threshold	Comorbidity	PA renewal	Step therapy	Out-of-pocket cost	Mean SE, smallest contrast
30	0.915	0.98	0.500	0.96	1	0.170
50	0.990	1.00	0.745	1.00	1	0.129
75	1.000	1.00	0.920	1.00	1	0.106
100	1.000	1.00	0.970	1.00	1	0.091
150	1.000	1.00	1.000	1.00	1	0.074
200	1.000	1.00	1.000	1.00	1	0.064
300	1.000	1.00	1.000	1.00	1	0.052

Power passes 80% for every attribute by about 75 respondents



Source: simulation (*research_pack/dce_design.R*) with illustrative true preference weights (an assumption), 200 replications per sample size.

Recommendation: at least **100 respondents** for pooled main effects (above the rule-of-thumb minimum of 63 and above the 75 where simulated power passes 80% for every attribute), **300** for a study with three prescriber subgroups of 100, and **600** if the three subgroups are to be compared with 200 each (Sawtooth's subgroup rule). The weakest attribute in simulation is the PA renewal interval, which has the smallest assumed effect.

6. Analysis framework and reporting outline

Step	Method
Data cleaning	Remove speeders (under one third of median completion time), straight-liners (same position in all 12 tasks), failed attention checks
Main model	Conditional logit with effects-coded attributes; mixed logit as a robustness check for preference heterogeneity
Outputs	Part-worth utilities with 95% CIs; relative attribute importance; willingness to trade off (here in out-of-pocket dollars)
Segments	Latent-class or covariate interactions by prescriber type and by state coverage history; payers analyzed separately
Simulation	Predicted preference share for candidate coverage policies (for example BMI 30 or more, 12-month renewal, no step therapy, \$0)
Reporting	Methods in one page; part-worth chart; trade-off table; limits (stated preference, hypothetical policies, small samples)

Outline of the final report: (1) executive summary and recommendations; (2) what we asked and whom; (3) how decisions are made today (qualitative); (4) trade-offs in coverage criteria (DCE); (5) implications by audience; (6) limits; (7) appendix: screeners, guide, stimulus, design.

7. Illustrative recommendation slides

! Important

Illustrative. No fieldwork has been done. These slides show the format the final report would use; where a number appears it is labeled *secondary data* and linked to its output, and every other statement is an assumption to be tested in the research.

Illustrative slide 1. Coverage works through managed care, so test the criteria that managed-care organizations apply

- *Secondary data* (Module C): coverage was associated with an estimated 12.2 additional Wegovy and Zepbound prescriptions per 1,000 Medicaid enrollees per quarter on average (95% CI 8.4 to 16.0), from 10 covering states versus 34 never-covering jurisdictions ([analysis/outputs/tables/overall_att.csv](#), row 1).
- Coverage operates mainly through managed-care programs in South Carolina and Rhode Island ([deviations.md](#) item 6).
- *Hypothesis for the research*: managed-care medical directors weigh the renewal interval and step therapy more than the BMI threshold.

Illustrative slide 2. Coverage can end quickly; ask what would sustain it

- *Secondary data* (Module C, preliminary, descriptive): after coverage ended, prescriptions per enrollee fell by -67.2% in California and -90.8% in Pennsylvania from 2025 Q4 to 2026 Q1 ([analysis/outputs/tables/withdrawal_change_2025Q4_to_2026Q1.csv](#)).
- *Hypothesis for the research*: budget predictability, not clinical doubt, drives the decision to end coverage.

Illustrative slide 3. The budget is the lever payers describe; test which assumptions they accept

- *Secondary data* (Module E): five-year net cost of 161.3 million dollars (2.69 dollars per enrollee per month) for a program of 1 million enrollees in the central case, a range of 55.6 to 369.4 million across scenarios, with the rebate assumed at 51.2% (midpoint of 23.1% and 79.3%) because actual net prices are confidential ([analysis/outputs/tables/moduleE_scenario_results.csv](#)).
- *Hypothesis for the research*: payers would accept tighter criteria in exchange for a predictable price; test through the cost and renewal attributes in the choice experiment.

8. Ethics, data and limits

No personal data are collected in this pack. A real study would need informed consent, screener data minimization, ethics or institutional review as applicable, and compensation set at fair-market rates without inducement. The design uses zero priors and assumed effect sizes that a pilot should replace.